

Galls Get the Gals

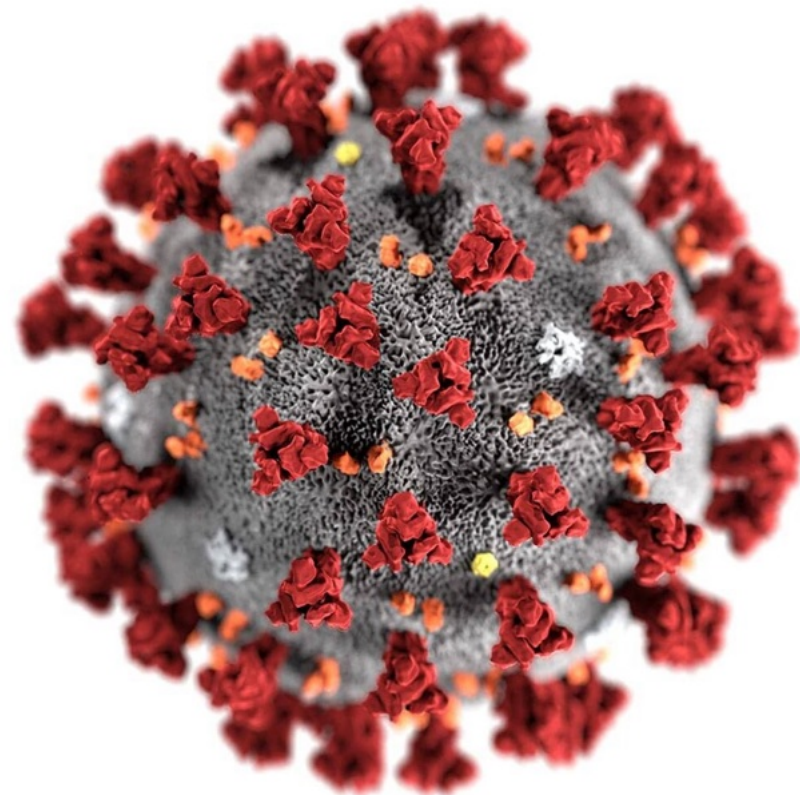
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Who am I, and why am I here?

I am a software engineer, but I have always been fascinated by nature. A confluence of events led me to galls...



iNaturalist

What the heck is a gall?

Plant galls are abnormal growths of plant tissue, similar to tumors or warts in animals, caused by insects, mites, fungi, bacteria, or other organisms. They are often intricate structures that can identify the species that made them, even when that organism is not visible.

The gall is not just damage; it is a structured outcome of the insect-plant interaction.

The plant is expressing the phenotype (observable traits) of the gall-former's genotype (DNA blueprint).

In other words...

This tiny wasp
~ 3.5 mm



forces this oak



to grow this



as a fully stocked larder and nursery for its offspring

Diversity of Gall-formers

- Cecidomyiidae (gall midges) a family of flies that has ~7000 sp.
- Eriophyidae (gall mites) ~3500 sp. many of which form galls
- Hemiptera (true bugs) several families produce galls
- Fungi, bacteria, nematodes, viruses, and even other plants can cause galls

Some argue that there are more than 200,000+ sp. of gall-forming insects (nearly all of which are undescribed)!

And one more very important family of gall-formers...

Cynipidae

A family of tiny (2-5mm) wasps

- ~1400 sp. worldwide.
 - Most are oak specialists, ~1000 sp.
 - Oak is most diverse in NA and thus most Oak Gall Wasp sp. are in NA, ~800 sp.
- Some can produce offspring via **pathogenesis**
- Some create 2 types of galls on different plant parts
- Some are **inquilines** they no longer make galls, they take over other's



Knowing the Host Plant is Critical

Many gall-formers are host-specific, if not to the species then to the genus. Thus host plant identification is critical to identifying a gall.

Even if you cannot identify the plant fully, document it as carefully as possible.

fsus.ncbg.unc.edu | floraofvirginia.org | iNaturalist.org

Common Questions

Do they hurt the plant?

Not usually. They are parasites, meaning the host survives; they are not parasitoids, which kill the host.

Is this a gall?

Not every bump, blister, or swelling is a gall.

How Can You Learn More?



gallformers

www.gallformers.org

Thank You!

Demo Time

**What About the Title of this
Presentation?**

Galls Get the Gals

Links:

Gallformers: <https://www.gallformers.org/>

Gallformers is an open source project and all source code for the site can be found here:
<https://github.com/jeffdc/gallformers>

Donate to gallformers: <https://www.patreon.com/gallformers>

Jeff's iNaturalist Profile: <https://www.inaturalist.org/people/jeffdc>

Oak Compendium is Jeff's Oak site, is very much in an incomplete state: <https://oakcompendium.org/>

Links to all photos:

Photo licenses and credits can be found at the links

Slide 4 L to R:

- Adult Callirhytis seminator: <https://www.inaturalist.org/observations/217143861>
- Quercus alba: <https://www.inaturalist.org/observations/118111221>
- Callirhytis seminator gall: <https://www.inaturalist.org/observations/156264973>

Slide 7 clockwise from top left:

- Atrusca centricola gall: <https://www.inaturalist.org/observations/35338661>
- Acraspis erinacei: <https://www.inaturalist.org/observations/178636125>
- Antron douglasii gall: <https://www.inaturalist.org/observations/307558669>
- Callirhytis tubicola: <https://www.inaturalist.org/observations/30756251>
- Andricus coronas: <https://www.inaturalist.org/observations/74986223>
- Feron kingi adult emerging from gall: <https://www.inaturalist.org/observations/265569046>
- Callirhytis gallastriatae: <https://www.inaturalist.org/observations/96982552>